Kenny Gruchalla, Ph.D.
Curriculum Vitae
(August 2, 2026)

Personal Information

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Experience

National Laboratory of the Rockies (formerly NREL), Golden, CO (June 2009 - present)

Distinguished Member of Research Staff

Dec 2025 - present **Acting Group Manager**

Jul 2024 - present **Principal Scientist**

Oct 2019 - Mar 2020 **Acting Group Manager**

Jun 2009 - Jul 2024 **Senior Scientist**

I lead the design, development, and strategic direction of a high-visibility, state-of-the-art visualization and human-computer interaction (HCI) laboratory. This facility supports the analysis and communication of large, complex, multivariate data and functions as an institutional prototyping environment for advancing emerging computing, visualization, and HCI technologies. I manage a multi-million-dollar annual budget and multidisciplinary teams. I regularly collaborate with C-level executives and senior stakeholders to ensure that laboratory capabilities align with institutional priorities, emerging research opportunities, and industrial partnerships. As Principal Investigator, I have successfully led a portfolio exceeding $7 million in competitively funded research projects. I serve as both a technical manager and a hands-on contributor, remaining actively involved in visualization and HCI–centric design and development, spanning advanced visualization hardware and software systems, from immersive environments to custom interaction devices.

University of Colorado at Boulder, Boulder, CO (April 2001 - Mar 2006, May 2011 - present)

Mar 2025 - present **Assistant Professor Adjoint**

May 2011 - Mar 2025 **Assistant Professor Adjunct**,
Department of Computer Science
I provide mentorship to doctoral-level students in the field of scientific data visualization, supporting the development of gradute students' research skills, including project design, data analysis, software development, and effective visualization techniques.

Apr 2001 - Mar 2006 **Professional Research Assistant**,
CADSWES (Center for Advanced Decision Support for Water and Environmental Systems)
I worked in an interdisciplinary research center on the design and the development of a commercial graphically-based decision support software system, *RiverWare*, implementing object-oriented simulation, rule-based simulation, and linear optimization to model watershed physical processes, water ownership, and policy.

Jan 2004 - Aug 2004 **Professional Research Assistant**,
Department of Molecular, Cellular, and Developmental Biology
I collaborated on the design and development of a pilot study to investigate the added value of using immersive visualization as a molecular research tool.

Jul 2002 - Aug 2003 **Graduate Research Assistant**,
BP Center for Visualization
I designed and developed an interactive 3D immersive application capable of integrating geological, geophysical, reservoir and well data with drilling and platform planning in an immersive virtual environment.

Colorado State University, Fort Collins, CO (October 2021 - present)

Oct 2021 - present **Affiliate Faculty**,
Richardson Design Center
As an Advisor for the Richardson Design Center (RDC), I support the development and execution of their graduate certificate program in human-centered design thinking. I bring industry experience and technical expertise to my advising role, providing a valuable resource for faculty and students as they navigate the rapidly evolving field of VR.

Red Canyon Engineering, Denver, CO (June 2000 - January 2009)

Jun 2003 - Jun 2009 **Research Scientist / Principal Software Engineer**

Jun 2000 - Jun 2003 **Software Engineering Consultant**
I led new business proposals and was the PI of two NASA-funded projects — the Mars Flight Simulator and Lunar Base Simulator. In addition, I conducted comprehensive software architecture and algorithm reviews for the Mars Odyssey and Genesis spacecraft programs, ensuring technical accuracy and project success.

National Center for Atmospheric Research (NCAR), Boulder, CO (May 2006 - October 2008)

May 2006 - Oct 2008 **Visitor Appointment**
I collaborated on the design and development of VAPOR, a state-of-the-art volume visualization suite designed to interactively explore large-scale time-varying multivariate computational fluid dynamics (CFD) data.

Raytheon, Aurora, CO (July 1995 - February 2001)

Jul 2000 - Feb 2001 **Senior Analyst / Medical Officer**,
United States Antarctic Program
Raytheon Polar Services
I managed the data acquisition and visualization laboratory aboard the National Science Foundation's Antarctic research vessel, the Nathaniel B. Palmer, providing scientific support to NSF grantees. I also sailed as Medical Officer (EMT) aboard the Palmer.

Oct 1997 - Jun 2000 **Technical Software Lead**,
Space and Science Systems, Raytheon Systems Corporation
I designed and developed animated meteorological visualization tools for the Cape Canaveral and Vandenberg space lift ranges that included the development of both real-time and analysis visualization algorithms and image processing software for radar and satellite instrumentation. As the technical lead I served as technical mentor, providing technical guidance across projects and organizations.

Jul 1995 - Oct 1997 **Software Developer**,
Satellite Mission Management Organization, Hughes Space Systems *(purchased by Raytheon in 1997)*
I designed and developed a distributed satellite mission planning and scheduling software system that included interactive 2D computer graphic models of satellite and ground station resource allocation and 3D modeling tools used for satellite payload constraint analysis. I also helped design, develop, and maintain object-oriented class libraries designed for reuse and rapid development of satellite space and ground applications.

Brookhaven National Laboratory (BNL), Upton, NY (January 1994 - May 1994)

Jan 1994 - May 1994 **Science and Engineering Research Intern**,
Advanced Technology Division
I designed and developed interactive 2D visualizations of subsurface radioactive waste plumes created by a physically-based model of the breach, leach, and transport of radioactive waste material.

Education

2009 **Ph.D. Computer Science**, *University of Colorado at Boulder*, Boulder, CO
Thesis: Progressive Visualization-Driven Multivariate Feature Definition and Analysis
Advisor: Professor Elizabeth Bradley
GPA: 3.9/4.0

2003 **M.S. Computer Science**, *University of Colorado at Boulder*, Boulder, CO
Thesis: Immersive Well-Path Planning: Investigating the added value of immersive visualization
Advisor: Professor Clayton Lewis
GPA: 3.9/4.0

1995 **B.S. Computer Science**, *New Mexico Institute of Mining and Technology*, Socorro, NM
GPA: 3.5/4.0

Honors and Awards

Research

- IEEE Computer Society Distinguished Visitor, 2025 Class (2025-present)
- NLR/NREL Distinguished Member of Research Staff (2021-present)
- IEEE Computer Society Distinguished Contributor, 2024
- ACM/Eurographics High Performance Graphics (HPG) 2024 Chair's Award
- NREL December 2024 Employee Team of the Month
  (with J.D. Laurence-Chasen, G. Johnson, and E. Grant)
- APS DFD 2020 Gallery of Fluid Motion Award
- NREL 2018 Innovation & Technology Transfer Outstanding New Partnership Award
- NREL 2017 Innovation & Technology Transfer Outstanding Public Information Award
- NREL May 2016 Employee Team of the Month
  (with N. Brunhart-Lupo and M. Stock)
- NREL FY14 Staff Award – Outstanding Achievement
- NREL 2013 President's Award
- DOE ACSR SciDAC 2011 People's Choice Award
- DOE ACSR SciDAC 2010 Outstanding Achievement in Scientific Visualization
- Advanced Imaging Magazine 2005 Imaging Solutions of the Year
- IEEE Visualization 2004 Visualization Contest Second Place

***Industry** (selected)*

- Raytheon Space and Science Systems CHIP Award
- Raytheon Systems Company Outstanding Achievement Award
- Hughes Space Systems Team Achievement Award

Academic

- New Mexico Tech Regents' Scholarship (4 years full tutition)

Funding (selected)

(>$7M competitive awards as PI; >$53M total)

September 2026 — Agentic GeoAI for Precision Mineral Exploration (AGAPEX), **Co-PI**
DOE Genesis Mission / AMMPT

October 2025 — SCADA Alarm Actionable Dashboard (SAAD), **Co-PI**
DOE OE

August 2024 — EVision 2040: Visualizing a Path Toward Sustainable Mobility, **PI**
JOET

July 2024 — Immersive Visualization Architecture for the NYS Power Grid Digital Twin, **PI**
WFO-New York Power Authority

December 2023 — DEWA Energy Visualization Analysis, **co-PI**
WFO-Dubai Energy and Water Authority

August 2023 — Immersive Digital Twin Laboratory for Engineering Education, **PI**
WFO-Fort Lewis College

October 2020 — Collaborative Visualization and Analysis for the Control Room of the Future, **PI**
DOE/NREL LDRD

August 2019 — Cluster Perception in Scatterplots, **PI**
DOE/NREL LDRD/PI4

April 2016 — Advanced Energy Systems Design Architecture, **PI**
DOE/NREL LDRD

September 2014 — Immersive Computational Steering & Modeling, **PI**
DOE/NREL LDRD

September 2014 — Visualization and Simulation of a Manufacturing Line, **PI**
WFO-Abengoa

January 2013 — Integrated Energy Management and Analysis for the ESIF's Computational Systems, **Co-PI**
DOE/NREL LDRD

September 2012 — Novel Visualization and Analysis for Extreme-Scale Wind Turbine Array Simulations, **PI**
DOE/NREL LDRD

January 2009 — Lunar Base Simulator, **PI**
NASA Glenn Research Center

March 2005 MarsFlight – An Immersive and Interactive Mars Airplane Simulator, **Co-PI**
NASA Glenn Research Center

Publications

(1000+ citations, g-index:33, h-index:17, i10-index:28, i1-index:55)

Book Chapters: *(2)*

- E. Gould, S. Guerin, C. Smith, S. Smith, B. Bush and K. Gruchalla. Learning and Tracking Ad Hoc Fiducial Markers in Spatial Augmented Reality In J.G. Michopoulos, C.J.J. Paredis, D.W. Rosen, J.M. Vance (Eds.), *Advances in Computers and Information in Engineering Research Vol. 2* ASMEE Press June 2021.
- K. Gruchalla and N. Brunhart-Lupo. The Utility of Virtual Reality for Science and Engineering In W.R. Sherman (Ed.), *VR Developer Gems*, CRC Press, May 2019.

Refereed Papers: *(41)*

- H. Goldwyn, G. Johnson, C. Ibarra, L. Padilla, K. Gruchalla. Beyond the Post Hoc User Study: Modeling Visual Decision-Making with Active Inference. *IEEE Transactions on Visualization and Computer Graphics*, IEEE VIS, in press.
- S. Molnar, G. Johnson, K. Gruchalla, K. Potter. Opportunities and Challenges in the Visualization of Energy Scenarios for Decision-Making *Energy Visualization 2024*, November 2024.
- N. Brunhart-Lupo, K. Gruchalla, L. Williams, S. Elias. Situated Visualization of Photovoltaic Module Performance for Workforce Development. *Energy Visualization 2024*, November 2024.
- G. Johnson, S. Molnar, N. Brunhart-Lupo, K. Gruchalla. Architecture for Web-Based Visualization of Large-Scale Energy Domains. *Energy Visualization 2024*, November 2024.
- G. Johnson, K. Gruchalla, M. Ingram, N. Guba, R. Cruickshank, and S. Caruso. Vis-SAGA: Visual Analytics for Situational Awareness of Grid Anomalies *VizSec2023*, October 2023.
- S. Molnar, K. Gruchalla, S. Dong, and J. Tan. Visualization of the Oscillatory Dynamics of an Island Power System. *Energy Visualization 2023*, October 2023.
- I. Lyons-Galante, M. Karimzadeh, S. Molnar, G. Johnson, and K. Gruchalla. Alternatives to Contour Visualizations for Power Systems Data. *Energy Visualization 2023*, October 2023.
- N. Brunhart-Lupo and K. Gruchalla. Immersive Particle Advection: Through the Scales of Renewable Energy. *ACM PEARC23*, July 2023.
- K. Gruchalla, S. Molnar, and G. Johnson. Reevaluating Contour Visualizations for Power Systems Data. *IEEE Transactions on Smart Grid*, March 2023.
- A. Podder, K. Gruchalla, N. Brunhart-Lupo, S. Pless, M. Sica and P. Lacchin. Immersive Industrialized Construction Environments for Energy Efficiency Construction Workforce. *Frontiers in Virtual Reality*, March 2022.
- S. Molnar, E. Bradley, K. Gruchalla. Oscillatory Spreading and Inertia in Power Grids. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, December 2021.
- H. Sitaraman, N. Brunhart-Lupo, M. Henry de Frahan, S. Yellapantula, B. Perry, J. Rood, R. Grout, M. Day, R. Binyahib, and K. Gruchalla. Visualizations of direct fuel injection effects in a supersonic cavity flameholder. *Physical Review Fluids*, November 2021.
- K. Gruchalla, S. Raghupathi, N. Brunhart-Lupo. Structure Perception in 3D Point Clouds *ACM SAP 2021*, September 2021.
- N. Brunhart-Lupo, S. Yellapantula, K. Gruchalla, R. Grout. Visualization of Jet Impingement and Ignition in a Piston-cylinder Chamber. *Energy Visualization 2021*, June 2021.
- M. Whitlock, D. Albers Szafir, K. Gruchalla. HydrogenAR: Interactive Data-Driven Presentation of Dispenser Reliability. *ISMAR 2020*, November 2020.
- N. Brunhart-Lupo, B. Bush, K. Gruchalla, K. Potter, S. Smith. Collaborative Exploration of Scientific Datasets using Immersive and Statistical Visualization. *Improving Scientific Software. NCAR/TB-564+PROC*, August 2020.
- A.L. Figueroa-Acevedo, C.-H. Tsai, K. Gruchalla, Z. Claes, S. Foley, J. Bakke, J. Okullo, A.J. Prabhakar. Visualizing the impacts of renewable energy growth in the U.S. Midcontinent. *IEEE Open Access Journal of Power and Energy*, January 2020.
- B. Bugbee, B.W. Bush, K. Gruchalla, K. Potter, N. Brunhart-Lupo, V. Krishnan. Enabling Immersive Engagement in Energy System Models with Deep Learning. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, June 2019.

- P.A. Fleming, J. Annoni, L.A. Martinez, S. Raach, K. Gruchalla, A. Scholbrock, M. Churchfield, and J. Roadman. Investigation into the shape of a wake of a yawed full-scale turbine *Journal of Physics: Conference Series* volume 1037, June 2018.
- S. Molnar and K. Gruchalla. Visualizing Electrical Power Systems as Flow Fields. In proceedings of *Workshop for Visualisation in Environmental Sciences (EnvirVis 2018)*, June 2018.
- P.A. Fleming, J. Annoni, M. Churchfield, L. Martinez, K. Gruchalla, M.J Lawson, and P.J. Moriarty. A simulation study demonstrating the importance of large-scale trailing vortices in wake steering *Wind Energy Science*, volume 3, May 2018.
- P.A. Fleming, J. Annoni, M.J. Churchfield, L. Martinez, K.M. Gruchalla, M.J Lawson, and P.J. Moriarty. From wake steering to flow control *Wind Energy Science Discussions*, November 2017.
- K. Gruchalla, N. Brunhart-Lupo, K. Potter, J. Clyne. Contextual Compression of Large-Scale Wind Turbine Array Simulations. In proceedings of *Workshop for Data Reduction for Big Scientific Data (DRBSD-2)*, November 2017.
- B. Bush, N. Brunhart-Lupo, B. Bugbee, V. Krishnan, K. Potter, and K. Gruchalla. Coupling Visualization, Simulation, and Deep Learning for Ensemble Steering of Complex Energy Models. In proceedings of *Data Systems for Interactive Analysis DSIA'17*, October 2017.
- B. Bugbee, C. Phillips, H. Egan, R. Elmore, K. Gruchalla, and A. Purkayastha. Prediction and characterization of application power use in a high-performance computing environment. *Statistical Analysis and Data Mining: The ASA Data Science Journal* February, 2017.
- G.A. Ferguson, V. Vorotnikov, N. Wunder, J. Clark, K. Gruchalla, T. Bartholomew, D.J. Robichaud, and G.T. Beckham. Ab Initio Surface Phase Diagrams for Coadsorption of Aromatics and Hydrogen on the Pt(111) Surface. *The Journal of Physical Chemistry C*, November 2016.
- K. Gruchalla, J. Novacheck, A. Bloom. Visualization of the Eastern Renewable Generation Integration Study. In proceedings of *SC16*, Salt Lake City, UT, November 2016.
- C. Zhang, S. Santhanagopalan, M.J. Stock, N. Brunhart-Lupo, K. Gruchalla. Interpretation of Simultaneous Mechanical-Electrical-Thermal Failure in a Lithium-Ion Battery Module. In proceedings of *SC16*, Salt Lake City, UT, November 2016.
- D. Macumber, K. Gruchalla, N. Brunhart-Lupo, M. Gleason, J. Abbot-Whitley, J. Robertson, B. Polly, K. Fleming, M. Schott. City Scale Modeling with OpenStudio. *In proceedings of ASHRAE and IBPSA-USA SimBuild 2016*.
- N. Brunhart-Lupo, B. Bush, K. Gruchalla, S. Smith. Simulation Exploration through Immersive Parallel Planes. In proceedings of *IEEE Workshop on Immersive Analytics,* March 2016.
- J. Hinkle, P.N. Ciesielski, K. Gruchalla, K.R. Munch, B.S. Donohoe. Biomass accessibility analysis using electron tomography. *Biotechnology for Biofuels* volume 8, November 2015.
- S. Li, K. Gruchalla, K. Potter, J. Clyne, and H. Childs. Evaluating the Efficacy of Wavelet Configurations on Turbulent-Flow Data. in *In proceedings of IEEE Symposium on Large Data Analysis and Visualization*, Chicago, IL, 2015.
- M. Lunacek, A. Nag, D. Alber, K. Gruchalla, C.H. Chang, and P.A. Graf. Simulation characterization and optimization of metabolic models with the high-performance systems biology toolkit. *SIAM Journal on Scientific Computing*, volume 33, pages 3402-3424, 2011.
- K. Gruchalla, M. Rast, E. Bradley, and P. Mininni. Segmentation and visualization of multivariate features using feature-local distributions. In *Advances in Visual Computing*, volume 6938 of *Lecture Notes in Computer Science*, pages 619–628. Springer Berlin / Heidelberg, 2011.
- M.A. Sprague, P.J. Moriarty, M.J. Churchfield, K. Gruchalla, S. Lee, J.K. Lundquist, J. Michalakes, and A. Purkayastha. Computational modeling of wind-plant aerodynamics. In proceedings of *SciDAC 2011, Denver, CO*, 2011.
- M. Guy, P. Earle, C. Ostrum, K. Gruchalla, and S. Horvath. Integration and dissemination of citizen reported and seismically derived earthquake information via social network technologies. In *Advances in Intelligent Data Analysis IX*, volume 6065 of *Lecture Notes in Computer Science*, pages 42–53. Springer Berlin / Heidelberg, 2010.

- J. Clyne, K. Gruchalla, and M. Rast. VAPOR: Visual, Statistical, and Structural Analysis of Astrophysical Flows. In proceedings of *Numerical Modeling of Space Plasma Flows: Astronum-2009 (Astronomical Society of the Pacific Conference Series)*, volume 429, pages 323-329, 2010.
- K. Gruchalla, M. Rast, E. Bradley, J. Clyne, and P. Mininni. Visualization-driven structural and statistical analysis of turbulent flows. In *Advances in Intelligent Data Analysis VIII*, volume 5772 of *Lecture Notes in Computer Science*, pages 321–332. Springer Berlin / Heidelberg, 2009.
- K. Gruchalla, M. Dubin, J. Marbach, and E. Bradley. Immersive examination of the qualitative structure of biomolecules. In proceedings of *International Workshop on Qualitative Reasoning about Physical Systems*, pages 36–41, 2008.
- K. Gruchalla, J. Marbach, and M. Dubin. Porting legacy applications to immersive virtual environments - a case study. In proceedings of *International Conference on Computer Graphics Theory and Applications (GRAPP 2007)*, pages 179–184, 2007.
- K. Gruchalla. Immersive well-path editing: investigating the added value of immersion. In proceedings of *IEEE Virtual Reality, 2004.*, pages 157 – 164, March 2004.

***Non-refereed Papers:** (19)*

- S.R. Spurgeon, M. Abolhasani, F. Baddour, R.B. Comes, V.P. Dravid, H. Egan, P. Emami, R.W. Epps, D.M. Fébba, R. Gannon, E.A. Gaulding, A. Ghosh, K. Gruchalla, et al. Born-Qualified: An Autonomous Framework for Deploying Advanced Energy and Electronic Materials, *arXiv preprint arXiv:2605.00639*, May 2026.
- N. Brunhart-Lupo, T. Dashmunkh, S. Ghosh, K. Gruchalla, G. Krishnamoorthy, S.K. Sadanandad, T. Ghaod. Three-dimensional Grid Visualization for Planning Activities: A Dubai Case Study, *NLR Technical Report NLR/6A40-98585*, July 2026.
- K. Gruchalla. Immersive Digital Twin Laboratory for Engineering Education: Cooperative Research and Development Final Report, *CRADA Number CRD-23-23838, NREL/TP-2C00-93161*, June 2025.
- S.L. Choi, R. Jain, P. Emami, K. Wadsack, F. Ding, H. Sun, K. Gruchalla, J. Hong, H. Zhang, X. Zhu, and B. Kroposki. eGridGPT: Trustworthy AI in the Control Room NREL Technical Report *NREL/TP-5D00-87740*, May 2024.
- I. Manogaran, A. Farthing, J. Maguire, K. Gruchalla. Savings in Action: Lessons Learned for a Vermont Community with Solar plus Storage NREL Technical Report *NREL/TP-7A40-84660*, January 2024.
- M. Ingram, A. Florita, M. Martin; K. Gruchalla, X. Fang, M. Cai, G. Johnson, N. Guba, A. Hasandka, M. Mooney, et al. Situational Awareness of Grid Anomalies (SAGA) for Visual Analytics NREL Technical Report No. *NREL/TP-5D00-85187*, February 2023.
- K. Gruchalla, G. Johnson. Heterogeneous Visualization: A Call for Empirically Driven Technology Advancement and Adoption. *ASCR Workshop on Visualization for Scientific Discovery, Decision-Making, and Communication*, January 2022
- R. King, A. Glaws, K. Gruchalla, M. Henry de Frahan. Extracting Maximum Information from Exascale Simulations *ASCR Workshop on Reimaging Codesign*, March 2021.
- M. Lawson, J. Melvin, S. Ananthan, K. Gruchalla, J. Rood, M. Sprague. Blade-Resolved, Single-Turbine Simulations Under Atmospheric Flow NREL Technical Report No. *NREL/TP-5000-72760*, January 2019.
- B. Palmintier, J. Giraldez, K. Gruchalla, P. Gotseff, A. Nagarajan, T. Harris, B. Bugbee, M. Baggu, J. Gantz, E. Boardman. Feeder Voltage Regulation with High Penetration PV using Advanced Inverters and a Distribution Management System: A Duke Energy Case Study NREL Technical Report No. *NREL/TP-5D00-65551*, November 2016.
- A. Bloom, A. Townsend, D. Palchak, J. Novacheck, J. King, C. Barrows, E. Ibanez, M. O'Connell, G. Jordan, B. Roberts, C. Draxl, K. Gruchalla. Eastern Renewable Generation Integration Study. NREL Technical Report No. *NREL/TP-6A20-64472*, August 2016.
  NREL 2017 Innovation & Technology Transfer Outstanding Public Information Award
- R. Elmore, K. Gruchalla, C. Phillips, A. Purkayastha, N. Wunder. An Analysis of Application Power and Schedule Composition in a High-Performance Computing Environment. NREL Technical Report No. *NREL/TP-2C00-65392*, January 2016.

- R.W. Grout, K. Malhorta, P. Ciesielski, K. Gruchalla, B. Donohoe, M. Nimlos. Computational Assessment of the effect of realistic intraparticle geometry on biomass heating rates and pyrolysis yields. *8th US National Combustion Meeting*, May 2013.
- G. Pech, K. Gruchalla, and J. Marbach. The case for visualization. *Exploration & Production*, January 2009.
- G.A. Dorn, G.S. Pech, K. Gruchalla, J. Marbach The Value of Visualization in Exploration and Production: Anecdotal Evidence and Quantitative Data *70th EAGE Conference & Exhibition-Workshops and Fieldtrips* June 2008.
- K. Gruchalla and J. Marbach. Interactively exploring multiple characteristics of hurricane simulation data. *Advanced Imaging*, 22, 2005.
- G. Dorn, K. Gruchalla, J. Carlson, J. Marbach, T. Southren, and A. Jamieson. A visualization-driven paradigm for adaptive well-path planning. In *Offshore Technology Conference*, 2004.
- K. Gruchalla and E. Joynt. Late Cretaceous and Cenozoic Reconstructions of the Southwest Pacific, Data Report NBP00-07B. United States Antarctic Program, 2000.
- K. Bliss, K. Gruchalla, and K. Gavahan. OBS Refraction Profiling for Crustal Structure in Bransfield Strait, Data Report NBP00-07A. United States Antarctic Program, 2000.

***Video** (selected):*

- N. Brunhart-Lupo, M.T. Henry de Frahan, L. Cheung, N. deVelder, G. Yalla, P. Mohan, J. Rood, G. Vijayakumar, K. Gruchalla, M.A. Sprague. ExaWind: Revealing the dynamics of wind turbines with high fidelity simulations. *APS Division of Fluid Dynamics Gallery of Fluid Motion*, Houston, TX, 2025.
- H. Sitaraman, M. Henry De Frahan, S. Yellapantula, B. Perry, J. Rood, R. Grout, M. Day, N. Brunhart-Lupo, R. Binyahib, K. Gruchalla. Direct fuel injection effects in a supersonic cavity flameholder *APS Division of Fluid Dynamics Gallery of Fluid Motion*, Chicago, IL, 2020.
  Gallery Award
- S. Witter Hicks, M. Churchfield, K. Gruchalla. Visualization of a Simulated LiDAR-Based Wind Turbine Wake Measurement Campaign, *SuperComputing 2016 (SC16), Salt Lake City, UT, November, 2016.*
- K. Gruchalla, M.J. Churchfield, P.J. Moriarty, , S. Lee S. Li, J.K. Lundquist, J. Michalakes, A. Purkayastha, and M.A. Sprague. Computational modeling of turbine-wake effects. *SciDAC 2011*, Denver, CO, 2011.
  Awarded SciDAC 2011 People's Choice Award
- K. Gruchalla, O. Desjardins, P. Pepiot, and A. Purkayastha. Numerical simulation of a turbulent liquid jet. *SuperComputing 2010 (SC10)*, New Orleans, LA, 2010.
- K. Gruchalla, M.J. Churchfield, P.J. Moriarty, and L. Martinez. Eddy simulation of wind farm / atmospheric boundary layer interaction. *SuperComputing 2010 (SC10)*, New Orleans, LA, 2010.
- K. Gruchalla, P. Pepiot, and O. Desjardins. Particle dynamics in a fluidized bed reactor. *SciDAC 2010*, Chattanooga, TN, 2010.
  Awarded SciDAC 2010 Outstanding Achievement in Scientific Visualization
- K. Gruchalla and J. Marbach. Atmosv: Immersive visualization of the hurricane Isabel dataset. *IEEE Visualization 2004*, Austin, TX, 2004.
  Awarded second place in the 2004 IEEE Visualization Contest

***Published Data** (4):*

- I. Lyons-Galante, M. Karimzadeh, and K. Gruchalla. Alternatives to Contour Visualization for Power Systems Data. https://github.com/isaiahlg/voltage-tiling, 2023.
- K. Gruchalla, S. Molnar, and G. Johnson. Visualization Encoding Experiments for Power Systems Analysis. National Renewable Energy Laboratory. https://data.nrel.gov/submissions/200, 2022.
- S. Raghupathi, N. Brunhart-Lupo, and K. Gruchalla. Caerbannog Point Clouds. National Renewable Energy Laboratory. 10.7799/1729892, 2020
- C. Phillips, K. Gruchalla, R. Elmore, A. Purkayastha, and N. Wunder. HPC Energy Research. National Renewable Energy Laboratory. https://cscdata.nrel.gov/#/datasets/d332818f-ef57-4189-ba1d-beea291886eb, 2017.

***Preregistrations** (2):*

- K. Gruchalla, G. Johnson, and S. Molnar. *Voltage Outlier Detection*. OSF Preregistration, https://osf.io/tmv6s, July 2022.

- K. Gruchalla, G. Johnson, and S. Molnar. *Frequency Distribution Analysis*. OSF Preregistration, https://osf.io/zn7ag, July 2022.

***Motion Picture Credits** (1):*

- K. Gruchalla. Computer Graphics Rendering, *Green China Rising*. National Geographic Film, 2012.

***Thesis** (2)*

- K. Gruchalla. *Progressive Visualization-Driven Multivariate Feature Definition and Analysis*. PhD thesis, University of Colorado at Boulder, 2009.
- K. Gruchalla. *Immersive well-path planning: The added value of immersive visualization.* Master's thesis, University of Colorado at Boulder, 2003.

***Posters** (selected):*

- M. Sprague, S. Ananthan, K. Gruchalla, M. Lawson, J. Rood, K. Swirydowicz, S. Thomas, G. Vijayakumar, P. Crozier, C.R. Dohrmann, J.J. Hu, A.B. Williams, J. Turner, A. Prokopenko, R. Moser, and J. Melvin. ExaWind: Exascale Predictive Wind Plant Flow Physics Modeling. Poster at *2019 Exascale Computing Project Annual Meeting*, Houston, TX. January 2019.
- B. Bugbee, C. Phillips, K. Gruchalla, R. Elmore, and A. Purkayastha. Exploring HPC Application Power Usage. Poster at *2016 Conference on Data Analysis (CODA),* Santa Fe, NM, 2016.
- J. Hinkle, P. Ciesielski, K. Gruchalla, B. Donohoe, and K. Munch. Segmentation of Lamellar Sheets of Cellulose using Electron Tomography. Poster at *2014 Annual BioEnergy Science Center (BESC) Retreat*, Chattanooga, TN, 2014.
- H. Scharf, K. Gruchalla, R. Elmore, N. Brunhart-Lupo, A. Purkayastha. Optimal Prioritized Compression using Wavelets for Analysis and Visualization of Extreme-Scale Wind Turbine Simulations. Poster at *2014 Conference on Data Analysis (CODA),* Santa Fe, NM, 2014.
- R. Elmore, M. Sheppy, N. Wunder, K. Munch, K. Gruchalla, A. Purkayastha. A Case Study in Demand-Response Strategies for Managing Power in HPC Environments. Poster at *2014 Conference on Data Analysis (CODA),* Santa Fe, NM, 2014.
- M. Dubin, A. Pardi, and K. Gruchalla. Using immersive virtual reality for visualization of macromolecules. Poster at *2004 Butcher Symposium on Genomics and Biotechnology,* Boulder, CO, 2004.
- K. Gruchalla and J. Marbach. Atmosv: Immersive visualization of the Hurricane Isabel dataset. Contest Entry at *IEEE Visualization 2004*, Austin, TX, 2004.
  Awarded second place in the 2004 IEEE Visualization Contest

***Press** (selected):*

- 38th Annual Gallery of Fluid Motion award winners announced, *AAAS EurekAlert!, November 2020.*
- The Man Behind the Goggles: Meet NREL's Visualization Guru, *NREL News, June 3, 2019.*
- R. Gold, Superpower: One Man's Quest to Transform American Energy. Simon & Schuster, June 2019.
- Decoding the Weather Machine. *PBS Nova*, Episode 7, Season 45, April 2018.
- Scientists Break This Virtual Power Grid to Save the Real One. *Popular Mechanics*, July/August 2015.
- Daley, D. Visualize the Future: Simulation helps NREL troubleshoot scenarios yet to happen. *Sound & Communications*, August 2014. (Cover article)
- Scanlon, W. Scientists go eye to eye with research at ESIF. *NREL News Feature*, July 2013. (http://www.nrel.gov/news/features/feature_detail.cfm/feature_id=2254)
- Pierce, E.R. Tour the National Renewable Energy Lab's Latest Research Center *energy.gov* June 2013. (http://energy.gov/articles/slideshow-tour-national-renewable-energy-lab-s-latest-research-center)
- Tucker, E. Supercomputing Drives Innovation. *Continuum Magazine*, (2), 6-9. 2012.
- Mosher. D. 10 Award-Winning Scientific Simulation Videos *Wired.com Wired Science* August 2011. (http://www.wired.com/wiredscience/2011/08/science-simulation-videos/)

Presentations (selected)

Keynotes *(3)*

- "Visualization for Insight and Data Analysis in Energy Research," Keynote address to TechX Natal26 Conference, Natal Brazil, May 2026.
- "Immersive Analytics: Knowledge Discovery from the Inside," Keynote Address to the 2016 CSU Virtual Reality Symposium, Fort Collins, CO, October 2016.
- "Scientific Data Analysis and Knowledge Discovery through Advanced Visualization," Keynote Address to the 2016 CAAV Conference, Denver, CO, June 2016.

Invited *(selected)*

- "Visualization for Insight and Data Analysis in Energy Research," IEEE Computer Society Schenectady Chapter Lecture, Schenectady, NY, July 2026.
- "The Digital Twin: Visualizing the Future of the Power Grid," IEEE Computer Society Grid Talk Series. IEEE Philippines Section. Manila, Philippines (virtual), May 2026.
- "Visualization for Insight, Analysis, and Decision Support in Energy Systems," NITRD-DOE-NSF DECIDE Workshop: Computational Decision Support for High-Consequence, Complex Problems, Falls Church, VA, March 2025.
- "Immersive Visualization for Scientific Data Analysis and Knowledge Discovery," Surveying VR Data Visualization Software for Research Seminar, Brown University, Providence, RI (virtual). January 2025.
- "Visualization within the Department of Energy: NREL," IEEE VIS Application Spotlight. IEEE VIS 2024, (virtual), October 2024
- "Immersive Visualization for Scientific Data Analysis," Immersive Visualisation for Science, Research, Art and Digital Twins Applications BOF, SIGGRAPH 2024, Denver, CO, August 2024
- "Advanced Visualization for Scientific Data Analysis and Insight," Computational Science Department Seminar, Princeton Plasma Physics Laboratory, Princeton, NJ (virtual), November 2023.
- "The Future of Visualization Technology," Visualization in Practice, IEEE VIS, Melbourne, Victoria, Australia, October 2023.
- "Visualization of Energy Systems," Monash University, Clayton, Victoria, Australia, October 2023.
- "From Data to Discovery at NREL," Computer Science and Engineering Seminar. New Mexico Tech, Socorro, NM. September 2023.
- "Advanced Visualization for Scientific Data Analysis," Summer Workshop on Energy Education for Teachers (SWEET). Colorado School of Mines, Golden, CO. June 2023.
- "Sustainable Immersive Visualization: The Tale of Two Visualization Labs," Workshop on Immersive Visualization Laboratories, IEEE VR 2023, Shanghai, China (virtual). March 2023.
- "Immersive Visualization for Scientific Data Analysis," VR Data Visualization Software for Research Seminar, Brown University, Providence, RI (virtual). January 2023.
- "Advanced Visualization Infrastructure for Scientific Data Analysis and Knowledge Discovery," International Computing in the Atmospheric Sciences (iCAS2022). Stresa, Italy. September 2022.
- "Embodied Cognition and Perception: Immersion for Scientific Data Analysis," The CAAV Speaker Series, Virtual, August 2022.
- "Visualization-Supported Energy Analysis for Resilient Cities," Workshop on Urban Visualisation: Energy Resilient Cities, Melbourne, Australia (virtual due to COVID). June 2020.
- "NREL Visualization Overview," Electronic Visualization Lab, University of Illinois, Chicago, IL. June 2019.
- "NREL Visualization Overview," SCI Institute, University of Utah, Salt Lake City, UT. May 2019.
- "NREL Visualization Overview," Emerging Analytics Center, University of Arkansas, Little Rock, AR. April 2019.
- "Computational Science," JeffCon 2019, Golden, CO, January 2019.

- “Immersive Analytics: Knowledge Discovery from the Inside,” LANL Information Science and Technology Institute (ISTI) Seminar, Los Alamos National Laboratory, NM, July 2016.
- “Visualization and Analysis of Large-Scale Wind Turbine Array Simulations,” Boulder Fluids Dynamics Seminar, Boulder, CO, February 2015.
- “Enabling Renewable Energy Research through Scientific Visualization,” Data Visualization Summit, Boston, MA, September 2013.
- “Computational Modeling and Visualization of Turbine-Wake Effects,” Frontiers in Computational Physics, Boulder, CO, December 2012.
- “Visualization-Driven Multivariate Feature Analysis using Feature-Local Distributions,” Colorado School of Mines, Joint Colloquia of AMS and EECS, Golden, CO, December 2011.
- “Enabling Renewable Energy Research through Scientific Visualization,” University of Colorado, Department of Computer Science Colloquium, Boulder, CO, March 2011.
- “Statistically Guided Multivariate Visualization and Analysis of Turbulence Structures,” National Renewable Energy Laboratory (NREL), Golden, CO, April 2009.
- “Interactive visualization and analysis of turbulence structures and their statistics,” Laboratory of Computational Dynamics Turb-Helio Seminar. Boulder, Colorado, February 2009.
- “Extending VAPOR‘s hardware-accelerated volume rendering capabilities,” Computational and Information Systems Laboratory (CISL) Seminar, National Center for Atmospheric Research (NCAR). Boulder, Colorado, August 2007.
- “Multivariate volume visualization,” Laboratory of Computational Dynamics Turb-Helio Seminar. Boulder, Colorado, November 2005.

Contributed *(selected)*

- “Situated Visualization of Photovoltaic Module Performance for Workforce Development,” EnergyVis 2024, IEEE VIS 2024, October 2023.
- “Vis-SAGA: Visual Analytics for Situational Awareness of Grid Anomalies,” IEEE Symposium on Visualization for Cyber Security (VizSec2023), IEEE VIS 2023, Melbourne, Victoria, Australia, October 2023.
- “Visualization of the Oscillatory Dynamics of an Island Power System.” EnergyVis 2023, IEEE VIS 2023, Melbourne, Victoria, Australia, October 2023.
- “Alternatives to Contour Visualizations for Power Systems Data,” EnergyVis 2023, IEEE VIS 2023, Melbourne, Victoria, Australia, October 2023.
- “Reevaulating Heatmaps for Power Systems Visualization,” *2023 DOE Computer Graphics Forum*, Idaho National Laboratory, Idaho Falls, ID, April 2023.
- “Immersive Digital Twins and Decision Making,” *2023 DICE Digital Engineering Conference*, Idaho National Laboratory, Idaho Falls, ID, April 2023.
- “Finding the Killer Rabbit of Caerbannog,” *2022 DOE Computer Graphics Forum*, Eugene, OR (virtual), September 2022.
- “Structure Perception in 3D Point Clouds,” *ACM Symposium on Applied Perception*, Paris, France (virutal), September 2021.
- “Knowledge Discovery through Immersive Analytics,” *High Performance Computing Symposium 2021*, Boulder, CO, May 2021.
- “Visualization Paradigms in the Renewable Energy Space,” *IEEE VIS 2019*, Application Spotlight, Vancouver BC, Canada, October 2019
- “Visualizing Electric Power Systems as Flow Fields,” *2019 DOE Computer Graphics Forum*, Monterey, CA, March 2019
- “Contextual Compression of Large-Scale Wind Turbine Array Simulations,” *Workshop for Data Reduction for Big Scientific Data, SC17*, Denver, CO, November 2017
- “Coupling Visualization, Simulation, and Deep Learning for Ensemble Steering of Complex Energy Models,” *DSIA: Data Systems for Interactive Analysis*, Phoenix, AZ, October 2017
- “Visualization of a Simulated LiDAR-Based Wind Turbine Wake Measurement Campaign,” *RMACC High Performance Computing Symposium 2017*, Boulder, CO, August 2017
- “Interpretation of Simultaneous Mechanical-Electrical-Thermal Failure in a Lithium-Ion Battery Module,” *SuperComputing 2016 (SC16)*, Salt Lake City, UT, November 2016

- "Visualization of the Eastern Renewable Generation Integration Study.," *SuperComputing 2016 (SC16)*, Salt Lake City, UT, November 2016
- "Virtual Reality Panel," *2016 CSU Virtual Reality Symposium*, Fort Collins, CO, October 2016
- "Transmission Grid Visualization," *Grid Modernization Laboratory Consortium: Tools and Data for Production Cost Modeling Workshop*, Golden, CO, October 2016.
- "ESIF Insight Center: Visualization Hardware Panel," *2013 DOE Computer Graphics Forum*, Portland, OR, April 2013
- "Quantifying and Meshing Features in Microscopy Data," *2012 NREL High-Performance Computing Workshop*, Golden, CO, August, 2012
- "Visual Analysis of Fluidized Bed Reactor Dynamics," DOE Computer Graphics Forum, Albuquerque, NM, April 2012.
- "Segmentation and visualization of multivariate features using feature-local distributions," *International Symposium on Visual Computing 2011*, Las Vegas, NV, September, 2011
- "The Dawn of Scientific Visualization at NREL," *2010 DOE Computer Graphics Forum*, Park City, Utah, April 2010.
- "Visualization-Driven Structural and Statistical Analysis of Turbulent Flows," *2009 Intelligent Data Analysis Conference*, Lyon, France, September 2009.
- "Immersive Examination of the Qualitative Structure of Biomolecules," *2008 International Workshop on Qualitative Reasoning about Physical Systems*, Boulder, Colorado, June 2008.
- "Porting legacy applications to immersive virtual environments – a case study," *The 2007 International Conference on Computer Graphics Theory and Applications*, Barcelona, Spain, March 2007.
- "Hardware-accelerated visualization of non-uniformly gridded volume data," *2007 DOE Computer Graphics Forum*, Peaceful Valley, Colorado, May 2007.
- "Accounting network visualization," 2005 Annual RiverWare User Group Meeting, Boulder, Colorado, March 2005.
- "Optimization and rules policy editor," 2005 Annual RiverWare User Group Meeting, Boulder, Colorado, March 2005.
- "Workspace migration to Qt," 2005 Annual RiverWare User Group Meeting, Boulder, Colorado, March 2005.
- "Immersive well-path editing: Investigating the added value of immersion," *IEEE Virtual Reality 2004 Conference*, Chicago, Illinois, March 2004.
- "The COE Kansas City flood control method," 2004 Annual RiverWare User Group Meeting, Boulder, Colorado, March 2004.
- "The added value of immersive visualization," 2004 Drilling Visualization Research Consortium Meeting, Boulder, Colorado, January 2004.
- "Corps of Engineers flood control methods," 2003 Annual RiverWare User Group Meeting, Boulder, Colorado, June 2003.
- "Plotting simulation data," 2001 Annual RiverWare User Group Meeting, Boulder, Colorado, June 2001.
- "Visualization and user interface development for breach, leach, and transport models," Brookhaven National Laboratory SERS Seminar. Upton, NY, May 1994

Students

Ph.D. Advising

Samantha Molnar, University of Colorado, Aug 2015 - Dec 2021
Inertia, Network Structure, and Power System Oscillations
Outcome: Hired as NREL Staff Scientist

Postdoc Advising

Nicholas Brunhart-Lupo, NREL, Nov 2014 - Oct 2016
Outcome: Hired as NREL Staff Scientist

Roba Binyahib, NREL, Apr 2020 - July 2021
Outcome: Hired as Intel Graphics Engineer

Ph.D. Committee

Sandra Bae, University of Colorado, March 2025
Advancing Interactive Physicalizations

	Matthew Whitlock, University of Colorado, November 2021 *Immersive Augmented Reality for Data-Driven Workflows*
	Nicholas Brunhart-Lupo, Colorado School of Mines, Oct 2014 *Morse decompositions of three-dimensional piecewise constant vector fields*
External Examiner	Cian Kelly, University College of Dublin, June 2025 *Diagramming Bulk Electrical Power Flows*
Interns	Sandra Bae, CU-Boulder (APUP), 2024-2025
	Isaiah Lyons-Galante, CU-Boulder (APUP), 2022-2024
	Sunand Raghupathi, Cornell (SULI), 2019
	Matthew Whitlock, CU-Boulder (RPP), 2019
	Samantha Molnar, CU-Boulder (APUP), 2016-2020
	Sean Yang, University of Washington (RPP), 2017
	Shane Witter Hicks, Principia College (SULI), 2016
	Nicholas Brunhart-Lupo, Colorado School of Mines (APUP), 2012-2014

Skills

Languages	C++, JavaScript, R, Python, Rust, Lisp, Fortran, OpenGL Shading Language, CUDA
Front-End	Qt/QML, D3.js, React, WebGL, deck.gl, Bevy, CSS
Libraries	OpenGL, VTK, ITK, OpenInventor, NetCDF, MPI
Software	VAPOR, ParaView, Avizo, JMP, LaTeX, Unity, Godot
Hardware	Immersive projection environments, high-resolution display walls, projection mapping, HPC, HMD (Apple Vision Pro, MagicLeap), 3D tracking systems
Skills	Technical management, software engineering, visualization design, research, technical writing, public speaking, proposal development, numerical algorithm development, GUI, data analysis
Research	Scientific visualization, immersive visualization, human-computer interaction, human factors, high-performance scientific computing, interactive computer graphics, and simulation

Professional Memberships

Senior Member	IEEE, IEEE Computer Society,
Member	IEEE Visualization and Graphics Technical Committee
Professional Member	Association for Computing Machinery (ACM)
Professional Member	ACM SIGGRAPH, ACM SIGCHI
Member	SIAM, 2006-2010
Member	Sigma Xi, 2006-2010
Professional Member	American Institute of Aeronautics and Astronautics (AIAA), 2003-2011

Service

Board Member	Computer Science & Engineering Advisory Board, New Mexico Tech, 2023-present
Steering Committee	DOECGF 2016 - present
Chair	EnergyVis 2024, St. Pete Beach, FL
Local Chair	High Performance Graphics 2024 (HPG24), Denver, CO
Chair	EnergyVis 2023, Melbourne, Australia
Co-Chair	Data Challenges in Sustainable Urban Energy, October 2023, Monash University, Australia
Chair	Visualization Paradigms in Renewable Energy, *IEEE VIS 2019*, Vancouver, Canada
Co-chair	EnergyVis 2021, Torino, Italy
Program Chair	DOECGF 2018, Savannah, GA
Site Chair	DOECGF 2017, Golden, CO
Program Committee	SimAUD 2017 - 2021
Program Committee	CityVis 2020
Program Committee	Intelligent Data Analysis 2011-2016
Mentor	IEEE VR Doctoral Consortium, 2021

Coach	Maple Grove Math Olympiads Team, 2018-2019, Golden, CO
Judge	CSU Virtual Reality Hackathon, 2016, Fort Collins, CO
Instructor	Maple Grove Minecraft Club, 2016-2017, Golden, CO
Judge	Korea Institute of Science and Technology Information Visualization Competition, 2014
Research Diver	Pacific Whale Foundation, Summer 1998, Wailuku HI
Reviewing	IEEE TVCG, IEEE VIS, LDAV, SIGCHI, IEEE VR, IEEE Access, PRESENCE, EuroVis, PacificVis, IDA, SimAUD, Journal of Information Science and Engineering
	R Journal, IEEE CG&A, IEEE TSM, Journal Chemometrics, Journal Environmental Science–Atomspheres, DOE ASCR
DOE Workshops	Autonomous Research for Real-World Science (ARROWS), 2025
	NITRD/DOE/NSF DECIDE Workshop:Computational Decision Support for High-Consequence, Complex Problems, 2025
	ASCR Workshop on Visualization for Scientific Discovery, Decision-Making, and Communication, 2022
	Computational Challenges in Energy Systems Integration and Grid Modernization, 2015
	DOE Exascale Workshop on Data Analysis, Management, and Visualization, 2011
	Workshop on Scientific Data Management and Informatics, 2009
Graduate	*Graduate Representative*, 2005-2006 Faculty Search Committee, CU Boulder
Undergraduate	*Member*, Solar Car Team, 1993-1994, NMT
	Senator, Student Senate, 1991-1992, NMT
	Member, Student Judiciary Board, 1990-1991, NMT

Open-Source Development

Tephrite, (Rust, Bevy) https://github.com/InsightCenterNoodles/tephrite_rs

Role: Design, Development

Tephrite is an adaptation of the Bevy graphics/game engine for immersive spaces. It consists of a collection of plugins that provide multiprocess scene replication, VRPN input handling, and common tools useful for building immersive 3D applications.

StatKeeper, (Swift). https://github.com/gruchalla/StatKeeper

Role: Sole Author

StatKeeper is a SwiftUI iPhone app for tracking a single basketball player's in-game performance.

travelmapr, (R) https://github.com/gruchalla/travelmapr

Role: Sole Author

travelmapr is an R package for visualizing personal travel histories.

Ziti, (Swift, RealityKit) https://github.com/InsightCenterNoodles/Ziti

Role: Design

Ziti is a NOODLES client application for RealityKit-powered devices (specifically the Apple Vision Pro).

PowerSystemsData Spec, (Cap'n Proto) https://github.com/InsightCenterNoodles/PowerSystemsData

Role: Development

The PSD specification defines a standardized data format for representing power system datasets, with a focus on supporting time-series visualizations and 3D visualization tools. This format is defined in Cap'n Proto.

FLC Solar Lab, (GDScript, Python) https://github.com/FLC-Solar-Lab

Role: Principal Investigator, Design, Development

Hardware and software infrastructure for integrating digital twin simulations with physical objects in a laboratory setting.

NOODLES, https://github.com/InsightCenterNoodles

Role: Design

NOODLES is a protocol specification for scenegraph synchronization, based off GLTF, using CDDL for the specification and CBOR as the message format.

3D Power Flow Visualization, (C++, OpenGL) https://github.nrel.gov/nbrunhar/Intellicamp3

Role: Design / Developer

An immersive software suite to visualize power flow simulations.

CatCost (Catalyst Cost Estimation Tool), (Javascript) catcost.chemcatbio.org

Role: Design

Cost estimation tool designed to reduce the cost uncertainty associated with pre-commercial catalyst materials development.

Surface Phase Explorer, (Javascript, WebGL) spe.nrel.gov

Role: WebGL Development

Understanding the coadsorption of two species on a surface requires the exploration of phase diagrams. To facilitate exploration this web tool creates interactive and downloadable phase diagrams from ab initio data.

kaleidescope, (R) github.com/NREL/kaleidoscope

Role: Sole Author

An R package developed to visualize PLEXOS scenarios.

VAPOR, (C++, OpenGL) www.vapor.ucar.edu

Role: Developer

Contributions: Volume rendering engines, transfer function interface, model parsing and rendering

An interactive 3D visualization and quantitative analysis software suite tailored towards terascale computational fluid dynamics data.

MarsFlight, (C++, OpenGL) education.grc.nasa.gov/MarsFlight

Role: Principal Engineer

Contributions: Terrain rendering, subsystems interface, map interface, and front-end configuration & deployment interface

An interactive flight simulator of a Mars airplane concept vehicle, which includes a complete model of the Martian terrain based on MOLA data and rover imagery. The flight simulator is based on the open source FlightGear (www.flightgear.org) project.

iPyMOL, (C++, OpenGL) contact dubin@colorado.edu

Role: Principal Engineer

Contributions: Immersive port

An immersive port of the PyMOL (pymol.sourceforge.net) molecular visualization system, adding interactive visualization support for head-tracked, stereoscopic immersive virtual environments.

Other

Certifications	Wilderness EMT / EMT-B, *WMI/NREMT* Open Water SCUBA Diver, *PADI* Deep Learning Specialization, *Coursera DeepLearing.ai*
Hobbies	Rock climbing, mountaineering, skiing, additive manufacturing, photography, wood working
Development Portfolio	*(available upon request)*
Security Clearances	*(available upon request)*
References	*(available upon request)*